Topic 6 — Further Mechanics

6.1 Impulse and Momentum

Momentum (p) is the product of an object's mass and velocity. It is a vector quantity, meaning it has both magnitude and direction. The direction of momentum is the same as the direction of velocity.

p = mv (unit: kg m s^{-1} or N s)

Momentum is one of the most fundamental conserved quantities in physics. Unlike kinetic energy, momentum is always conserved in all types of collisions (provided no external forces act). The concept of momentum is central to understanding how forces change the motion of objects.

Impulse

Impulse is defined as the change in momentum of an object. It equals the product of the resultant force and the time for which it acts. Impulse is also a vector quantity, and its direction is the same as the direction of the resultant force.

$$\text{Impulse} = F\Delta t = \Delta p = mv - mu$$

The unit of impulse is N s (equivalent to kg m s^{-1}). This equivalence can be shown by dimensional analysis: 1 N s = 1 (kg m s^{-2})(s) = 1 kg m s^{-1}.

Newton's second law can be written in its more general form using impulse:

$$F = \Delta p / \Delta t$$

This form is more general than F = ma because it applies even when the mass of the object is changing (e.g., a rocket expelling fuel).

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Force-Time Graphs

On a force-time graph, the **area under the curve** equals the impulse (change in momentum). This applies whether the force is constant or varying.

- For a constant force, the area is a simple rectangle: impulse = $F \times \Delta t$
- For a linearly varying force, use the area of a triangle or trapezium
- For a variable force, estimate the area using counting squares or the trapezium rule

Worked Example:

A tennis ball of mass 58 g is travelling at 20 m s^{-1} towards a racket. After being hit, it travels at 30 m s^{-1} in the opposite direction. The contact time is 5.0 ms. Calculate the average force on the ball.

Solution: Taking the initial direction as positive:
$u = +20$ m s^{-1}, $v = -30$ m s^{-1} (reversed direction), $m = 0.058$ kg
$\Delta p = mv - mu = 0.058 \times (-30) - 0.058 \times (20) = -1.74 - 1.16 = -2.90$ N s
$F = \Delta p / \Delta t = -2.90 / 0.005 =$ **-580 N**
The magnitude of the average force is 580 N, directed away from the racket (in the direction the ball leaves).

Exam Tip: If a question gives you a force-time graph, calculate the area under it to find the impulse. Always include units (N s). Be careful with direction: if an object bounces back, the change in momentum is the sum of the magnitudes of the initial and final momenta (since the velocity reverses sign).

6.2 Conservation of Momentum

The principle of conservation of linear momentum: In a closed system (no external resultant force), the total momentum before an interaction equals the total momentum after the interaction.

This principle arises directly from Newton's third law. When two objects interact, they exert equal and opposite forces on each other for the same time. Therefore, the impulse on object A is equal and opposite to the impulse on object B, meaning the total momentum of the system is unchanged.

Worked Example:

A 2.0 kg trolley moving at 3.0 m s^{-1} collides with a stationary 1.0 kg trolley. After the collision, the 2.0 kg trolley moves at 1.0 m s^{-1} in the same direction. Find the velocity of the 1.0 kg trolley after the collision and determine whether the collision is elastic.

Solution:

Total momentum before = 2.0 × 3.0 + 1.0 × 0 = 6.0 kg m s^{-1}
Total momentum after = 2.0 × 1.0 + 1.0 × v
By conservation: 6.0 = 2.0 + v, so v = **4.0 m s^{-1}**

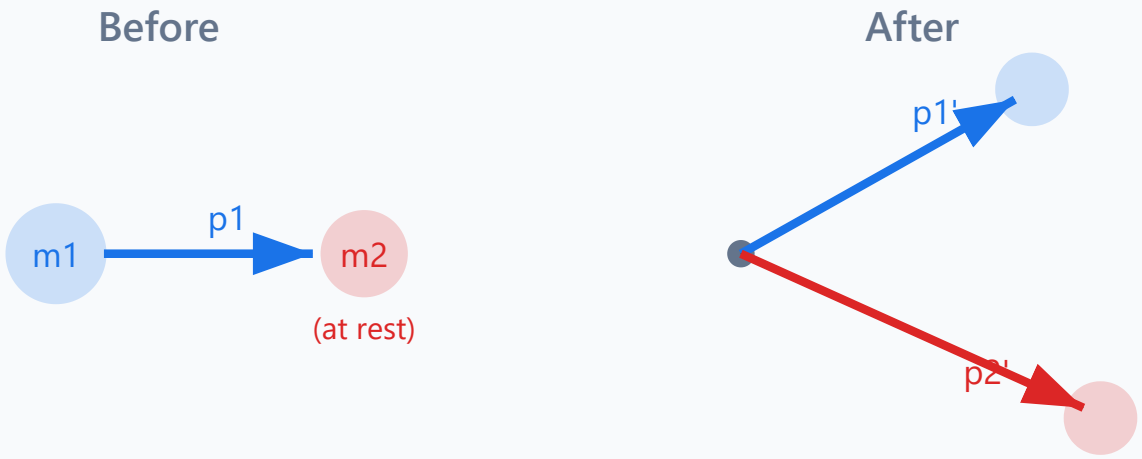
KE before = ½ × 2.0 × 3.0^2 = 9.0 J
KE after = ½ × 2.0 × 1.0^2 + ½ × 1.0 × 4.0^2 = 1.0 + 8.0 = 9.0 J
KE before = KE after, so the collision is **elastic**.

Conservation of Momentum in 2D

In two dimensions, momentum is conserved independently in each perpendicular direction. You resolve momentum into x and y components and apply conservation separately:

- Total x-momentum before = Total x-momentum after
- Total y-momentum before = Total y-momentum after

2D Collision: Momentum Conservation



x-momentum: $p1 = p1'\cos\theta1 + p2'\cos\theta2$

y-momentum: $0 = p1'\sin\theta1 - p2'\sin\theta2$

Worked Example:

A snooker ball of mass 0.16 kg moving at 2.0 m s^{-1} strikes a stationary ball of equal mass. After the collision, the first ball moves at 1.2 m s^{-1} at 40° to the original direction. Find the speed and direction of the second ball.

Solution:

x-direction: $0.16 \times 2.0 = 0.16 \times 1.2 \cos 40° + 0.16 \times v \cos \theta$

$0.32 = 0.147 + 0.16v \cos \theta$

$0.16v \cos \theta = 0.173$ ... (1)

y-direction: $0 = 0.16 \times 1.2 \sin 40° - 0.16 \times v \sin \theta$

$0.16v \sin \theta = 0.123$ ... (2)

Dividing (2) by (1): $\tan \theta = 0.123 / 0.173 = 0.714$, so θ = **35.5°**

From (1): $v = 0.173 / (0.16 \times \cos 35.5°)$ = **1.33 m s^{-1}**

Vector diagrams can be drawn to show that the total momentum vectors form a closed triangle (or polygon) if momentum is conserved.

Elastic vs Inelastic Collisions

Property	Elastic Collision	Inelastic Collision
Momentum	Conserved	Conserved
Kinetic Energy	Conserved	Not conserved (some KE converted to heat, sound, deformation)
Total Energy	Conserved	Conserved
Examples	Collisions between gas molecules, hard spheres at low speeds	Most real-world collisions, car crashes, catching a ball

A **perfectly inelastic** (totally inelastic) collision is one where the objects stick together after impact. Maximum kinetic energy is lost (consistent with momentum conservation). An **explosion** is the reverse of a perfectly inelastic collision: objects initially together move apart, with KE increasing (converted from chemical or other stored energy).

Exam Tip: To check if a collision is elastic, calculate the total KE before and after. If they are equal, it is elastic. Always show your working clearly. Remember: momentum is always conserved in collisions (provided no external forces act), but kinetic energy is only conserved in elastic collisions.

6.3 Circular Motion

Circular motion occurs when an object moves in a circular path at constant speed. Although speed is constant, the velocity is continuously changing direction, so the object is accelerating. This acceleration is always directed towards the centre of the circle.

Angular Velocity

Angular velocity (ω) is the rate of change of angular displacement. It is measured in radians per second (rad s^{-1}).

$$\omega = 2\pi / T = 2\pi f$$

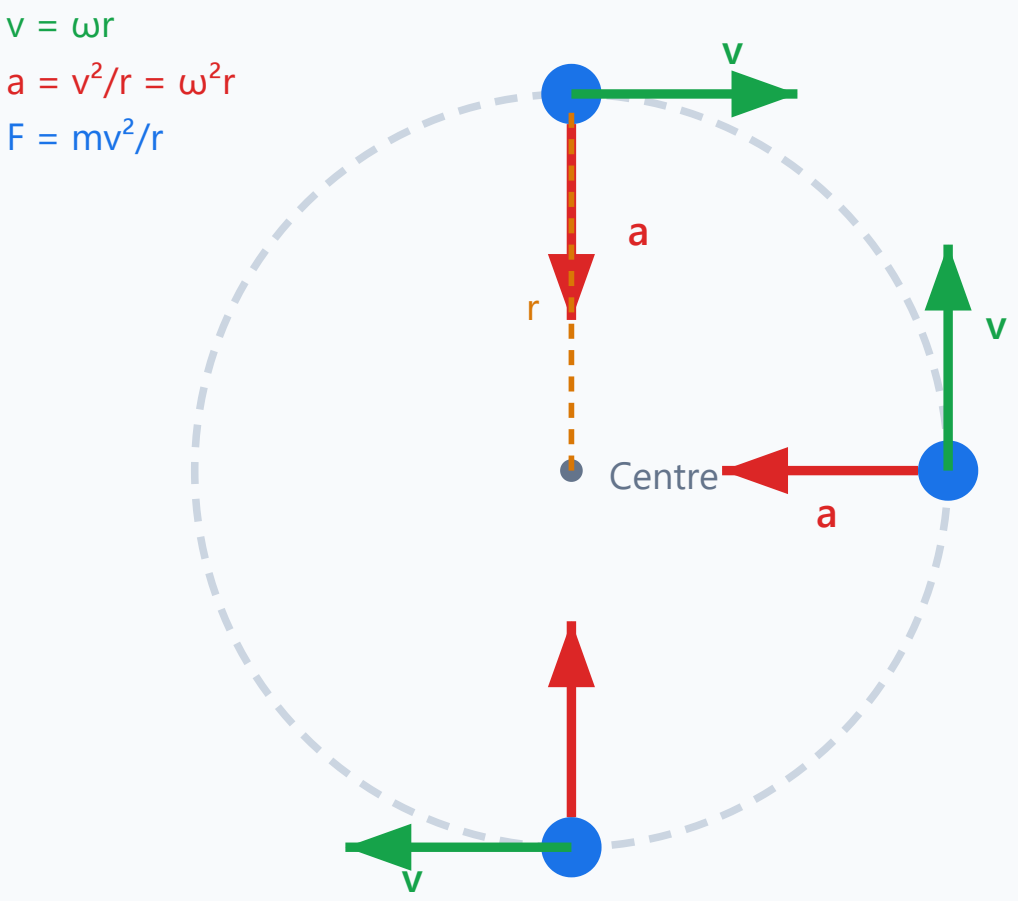
where T is the period (time for one complete revolution) and f is the frequency (number of revolutions per second).

The relationship between linear speed v and angular velocity:

$$v = \omega r$$

Note that one complete revolution = 2π radians = 360°. The radian is the SI unit of angle. 1 radian is the angle subtended at the centre of a circle by an arc equal in length to the radius.

Circular Motion: Forces and Vectors



v is always tangential | a always points to centre

Centripetal Acceleration

An object in uniform circular motion has an acceleration directed towards the centre of the circle. This is the **centripetal acceleration**. It arises because the velocity vector is constantly changing direction, even though its magnitude (speed) remains constant.

$$a = v^2 / r = \omega^2 r$$

The derivation uses the vector change in velocity over a small time interval. As the object moves through a small angle $\Delta\theta$, the change in velocity Δv has magnitude $v\Delta\theta$ and points towards the centre. Therefore $a = \Delta v/\Delta t = v(\Delta\theta/\Delta t) = v\omega = v^2/r$.

Centripetal Force

The **centripetal force** is the resultant force directed towards the centre of the circle that causes circular motion. It is not a new type of force -- it is provided by existing forces (tension, gravity, friction, normal reaction, etc.).

$$F = mv^2 / r = m\omega^2 r$$

Exam Tip: Never label a force as "centripetal force" on a free-body diagram. Instead, identify the real force providing the centripetal force (e.g., tension, weight, friction, normal contact force). The centripetal force is the resultant of the real forces directed towards the centre.

Examples of Circular Motion

Scenario	What provides the centripetal force
Car going around a flat bend	Friction between tyres and road
Satellite orbiting Earth	Gravitational force
Ball on a string (horizontal circle)	Horizontal component of tension
Electron orbiting a nucleus (classical model)	Electrostatic (Coulomb) force

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Clothes in a washing machine drum	Normal reaction from the drum wall
Banked track (no friction)	Horizontal component of normal reaction

Worked Example:

A car of mass 1200 kg travels around a flat circular bend of radius 50 m at a constant speed of 15 m s^{-1}. Calculate: (a) the centripetal acceleration, (b) the centripetal force, (c) the minimum coefficient of friction needed.

Solution:

(a) $a = v^2/r = 15^2/50 =$ **4.5 m s^{-2}**

(b) $F = ma = 1200 \times 4.5 =$ **5400 N**

(c) Friction provides the centripetal force: $\mu mg = mv^2/r$

$\mu = v^2/(rg) = 225/(50 \times 9.81) =$ **0.46**

Banked Tracks

On a banked track (no friction), the horizontal component of the normal reaction provides the centripetal force:

- Vertical: $N \cos \theta = mg$
- Horizontal: $N \sin \theta = mv^2/r$
- Dividing: $\tan \theta = v^2/(rg)$

The banking angle determines the speed at which no friction is needed to maintain circular motion. At speeds above or below this design speed, friction is needed to prevent the vehicle sliding outward or inward respectively.

Vertical Circles

For an object on a string in a vertical circle, the centripetal force equation must account for the weight component:

- **At the top:** $T + mg = mv^2/r$ (both tension and weight act towards centre)
- **At the bottom:** $T - mg = mv^2/r$ (tension acts towards centre, weight acts away)

- **At the side (3 o'clock):** $T = mv^2/r$ (weight is perpendicular to the radius, so does not contribute to centripetal force; it causes the speed to change instead)

The minimum speed at the top occurs when tension T = 0:

$$v_{min} = \sqrt{(rg)} \quad \text{(at the top)}$$

If the speed at the top drops below this value, the object will not complete the circle and will fall away from the circular path.

Worked Example:

A bucket of water is swung in a vertical circle of radius 0.80 m. What is the minimum speed at the top of the circle to prevent the water falling out?

Solution:

At the top, the water stays in the bucket when the centripetal acceleration equals or exceeds g:

$v_{min} = \sqrt{(rg)} = \sqrt{(0.80 \times 9.81)} = \sqrt{(7.85)} =$ **2.8 m s^{-1}**

Conical Pendulum

A conical pendulum consists of a mass on a string that traces a horizontal circle. The string makes angle θ with the vertical.

- Vertical equilibrium: $T \cos \theta = mg$
- Centripetal force: $T \sin \theta = m\omega^2 r$
- Since $r = L \sin \theta$ (where L is the string length), dividing gives: $\tan \theta = \omega^2 r / g$
- The period $T = 2\pi\sqrt{(L \cos \theta / g)}$

Quantity	Symbol	Unit
Angular velocity	ω	rad s^{-1}
Period	T	s
Frequency	f	Hz

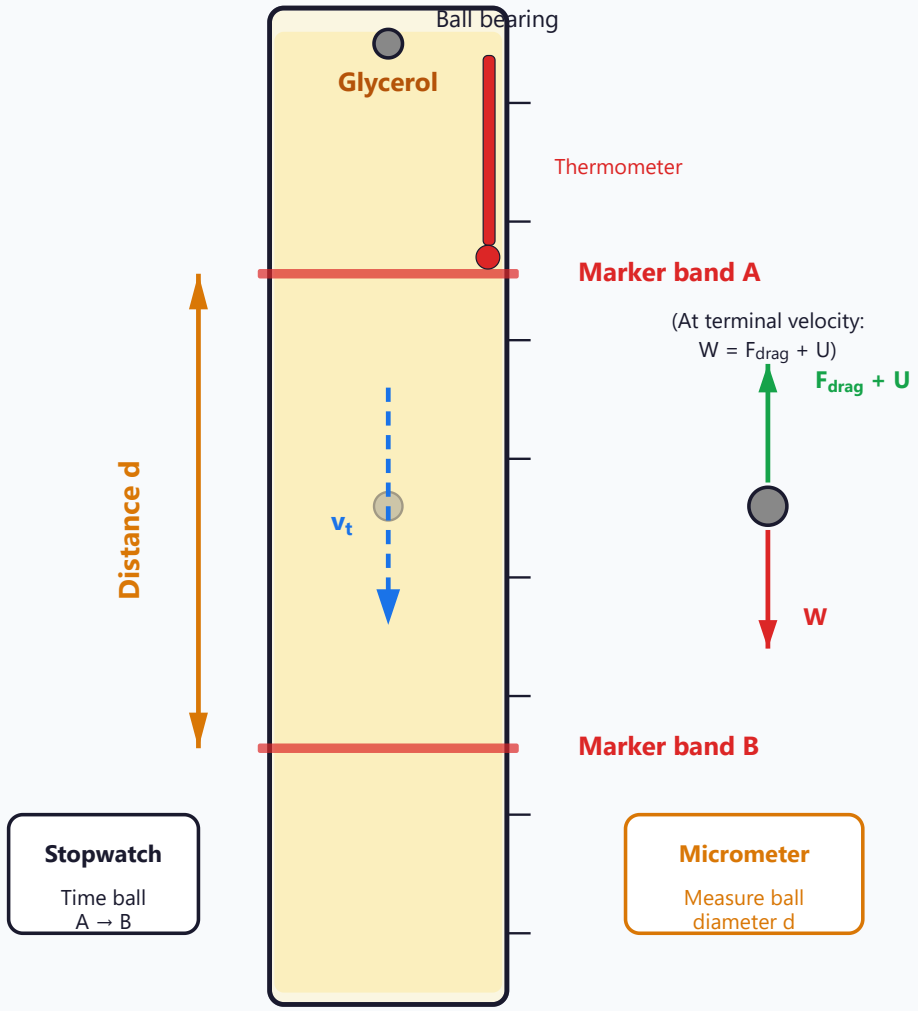
Centripetal acceleration	a	$m\ s^{-2}$
Centripetal force	F	N
Radius	r	m

Exam Tip: In vertical circle problems, always draw a free-body diagram at the point of interest. Apply Newton's second law towards the centre: net inward force = mv^2/r. At the top, both weight and tension act inward. At the bottom, tension acts inward but weight acts outward. The tension is greatest at the bottom and least at the top.

6.4 Key Relationships Summary

Concept	Key Equation	When to Use
Momentum	$p = mv$	All collision and interaction problems
Impulse	$F\Delta t = \Delta p$	When force and time are given, or force-time graph
Newton's 2nd law (general)	$F = \Delta p/\Delta t$	When mass changes (rockets, etc.)
Angular velocity	$\omega = 2\pi/T = 2\pi f$	Converting between period, frequency, and ω
Linear speed	$v = \omega r$	Relating linear and angular quantities
Centripetal acceleration	$a = v^2/r = \omega^2 r$	Finding acceleration in circular motion
Centripetal force	$F = mv^2/r = m\omega^2 r$	Finding the net inward force
Banking angle	$\tan\theta = v^2/(rg)$	Frictionless banked track problems
Min speed at top	$v = \sqrt{(rg)}$	Vertical circles (T = 0 at top)

🔬 Core Practical 11: Determine the viscosity of a liquid by the falling-ball method



CP11: Falling ball viscosity

(Stokes' law method)

Background Theory

Viscosity (η) is a measure of a fluid's resistance to flow. High-viscosity fluids (like glycerol, honey) resist motion more than low-viscosity fluids (like water). When a small sphere falls through a viscous fluid at low speed, the drag force is given by **Stokes' law**:

$$F_{drag} = 6\pi\eta rv$$

This applies only for **laminar (streamlined) flow** at low Reynolds numbers (Re < ~1). Laminar flow means the fluid moves in smooth layers without turbulence.

Apparatus

- Tall measuring cylinder (or tall glass tube) filled with viscous liquid (e.g. glycerol)
- Small steel ball bearings of various diameters (typically 1-5 mm)
- Micrometer screw gauge (to measure ball diameter, resolution ±0.01 mm)
- Rubber bands or tape as distance markers on the cylinder
- Stopwatch (resolution ±0.01 s)
- Metre ruler
- Thermometer
- Magnet (to retrieve ball bearings)
- Balance (to determine liquid density if needed)

Method

1. Fill the tall measuring cylinder with glycerol. Allow it to settle and reach a uniform temperature. Record the temperature (viscosity depends strongly on temperature).
2. Place two rubber band markers around the cylinder, well below the top (to allow the ball to reach terminal velocity before the first marker) and well above the base.
3. Measure the distance *d* between the two markers with a ruler.
4. Measure the diameter of the ball bearing using a **micrometer** (at least 3 readings in different orientations, taking the mean). Calculate the radius *r*.
5. Gently release the ball bearing centrally at the surface of the liquid. **Do not throw it.**
6. Start the stopwatch as the ball passes marker A. Stop it as the ball passes marker B. Record the time *t*.
7. Repeat at least 3 times with the same ball size and average the time.
8. Calculate terminal velocity: $v_t = d / t$.
9. Repeat with ball bearings of different diameters.

Safety

- Glycerol is slippery -- wipe up any spills immediately to prevent slipping.
- Wash hands after handling glycerol.
- Use a tall, stable cylinder to prevent it toppling. Consider clamping it to a retort stand.
- Use a magnet to retrieve steel ball bearings from the glycerol rather than reaching in.

Analysis

At terminal velocity, the forces on the ball are balanced:

$$\text{Weight} = \text{Drag force} + \text{Upthrust}$$

$$(4/3)\pi r^3 \rho_s g = 6\pi\eta r v_t + (4/3)\pi r^3 \rho_l g$$

Rearranging **Stokes' law**:

$$\eta = 2r^2(\rho_s - \rho_l)g \,/\, (9v_t)$$

Where η = viscosity (Pa s), ρ_s = density of ball, ρ_l = density of liquid, r = radius of ball, v_t = terminal velocity.

Plot $\mathbf{v_t}$ against $\mathbf{r^2}$ -- this should give a straight line through the origin with gradient = $2(\rho_s - \rho_l)g / (9\eta)$.

Worked Example:

A steel ball bearing (density 7800 kg m^{-3}) of diameter 3.0 mm falls through glycerol (density 1260 kg m^{-3}) and reaches a terminal velocity of 0.28 m s^{-1}. Calculate the viscosity of glycerol.

Solution:

$r = 1.5 \times 10^{-3}$ m

$\eta = 2r^2(\rho_s - \rho_l)g / (9v_t)$

$\eta = 2 \times (1.5 \times 10^{-3})^2 \times (7800 - 1260) \times 9.81 / (9 \times 0.28)$

$\eta = 2 \times 2.25 \times 10^{-6} \times 6540 \times 9.81 / 2.52$

η = **0.115 Pa s** (approximately 0.12 Pa s)

Sources of Error & Improvements

Source of Error	Type	Improvement
Ball may not have reached terminal velocity at marker A	Systematic	Place marker A well below the surface (at least 1/4 of cylinder height) to allow the ball to accelerate fully
Temperature changes affect viscosity significantly	Systematic	Record and maintain a constant temperature; perform experiment in a temperature-controlled environment
Ball may not fall centrally (wall effects / Faxen correction)	Systematic	Release the ball in the centre; use a wide cylinder (diameter ≥ 10× ball diameter)
Reaction time when using stopwatch	Random	Repeat timings and take the mean; use a longer distance between markers; consider video analysis
Ball not perfectly spherical	Random	Measure diameter in multiple orientations; reject any deformed balls
Parallax error when reading ball position against markers	Random	Position eyes at the same level as the marker; use a set square

Exam Tip: Stokes' law ($F = 6\pi\eta rv$) only applies for **laminar flow** at low Reynolds numbers. If the ball is too large or falls too fast, turbulence occurs and the drag force is no longer given

by Stokes' law. This is why small ball bearings in a very viscous liquid are used. You can verify terminal velocity has been reached by checking that the time for equal distances is constant.